

# Project Charter

Document 4 of 4 — AurumXchange Business Case

*Project title, SMART objectives, scope, deliverables, stakeholders, timeline, resources, risk register, and authorization.*

Prepared by Smart Innovations Ghana Limited

For Ghana Chamber of Gold Buyers

Date 12 August 2026

Classification Confidential — Board Distribution Only

This Project Charter formally authorizes the AurumXchange initiative, defines its execution boundaries, and grants the Project Manager authority to expend resources within the scope and budget defined herein. It is the document the Sponsor signs to release the CapEx funding; without a signed Charter, no personnel costs in Table 3.1 may be incurred. The Charter draws its business case from the BMC and its budget envelope from the Financial Documentation — every section below is traceable to one of those two documents, and the Golden Thread Validation section at the end of this business case confirms the traceability.

## 1. Project Title & Description

Project title: AurumXchange — A Regulated Digital Gold Trading Infrastructure for the Ghana Chamber of Gold Buyers. AurumXchange is a B2B SaaS platform that combines a member-verification portal, an RFQ-driven trading engine, partner-bank escrow, and an immutable compliance audit trail aligned to the LBMA Responsible Gold Guidance into a single regulated exchange. It will be built, hosted, and operated by Smart Innovations Ghana Limited under a 5-year exclusive revenue-sharing agreement with the Ghana Chamber of Gold Buyers, with the Chamber owning member relationships and governance and Smart Innovations owning the technology. The MVP scope (Phase 1 Digital Member Portal + Phase 2 Trading Platform) launches in Q3 2027; Phase 3 (Value-Added Services, including GoldBod and PMMC deep integration) and Phase 4 (Analytics) follow in 2028–2029. The platform is designed against Ghana's 2024 actuals — US\$11.64B in gold export revenue, 4.9M ounces of production, and the regulatory environment shaped by the GoldBod Act 1140 of 2 April 2025 [UNCTAD, 2025; CGTN Africa, 2025; trade.gov, 2025].

## 2. Business Case Summary

This project operationalizes the VPC Fit Statement ('For licensed gold buyers and sellers in Ghana who must trade under strict GoldBod, PMMC, Minerals Commission, and LBMA RGG compliance without exposing themselves to price opacity, counterparty risk, or funds-at-risk windows, AurumXchange is a regulated B2B gold trading infrastructure...'). The BMC confirms the commercial viability of that fit: hybrid SaaS monetization (0.10% transaction fees per side + tiered subscriptions + VAS) at a 70/30 Smart Innovations / Chamber revenue split. The Financial Documentation confirms the economic viability: CapEx of US\$463,450 (against a US\$450,000 ceiling — see Section 7.2 for the reconciliation), break-even at month 22 (Q3 2029 given a Q3 2027 launch), and 5-year cumulative net contribution of US\$4.05M to Smart Innovations Ghana Limited. The Charter authorizes the work, fixes scope, and assigns governance so that the financial projections in Document 3 can be executed rather than aspirational.

### 3. Objectives (SMART)

Objectives are written in SMART form — Specific, Measurable, Achievable, Relevant, Time-bound. They are the criteria against which the Charter's success will be judged at the post-launch 90-day review.

**Table 4.1. SMART Objectives**

| #   | Objective                             | Measurement                              | Target                                                                   | Deadline                    |
|-----|---------------------------------------|------------------------------------------|--------------------------------------------------------------------------|-----------------------------|
| O1  | MVP live and serving verified members | Production deployment + first real trade | ≥1 real gold trade settled via escrow                                    | 30 September 2027           |
| O2  | Verified member adoption              | Verified members onboarded               | 250 verified members                                                     | 30 September 2028 (Y1 end)  |
| O3  | Trade volume through platform         | Cumulative Year-1 trade value            | ≥ US\$350M                                                               | 30 September 2028           |
| O4  | System reliability                    | Uptime (monthly)                         | 99.9% (≤ 43 min downtime/month)                                          | Continuous, from launch     |
| O5  | Performance                           | P95 API response time                    | < 500ms for all critical endpoints                                       | Continuous, from launch     |
| O6  | Security posture                      | Critical-severity findings at launch     | 0 critical, ≤ 2 high (with remediation plan)                             | Pre-launch penetration test |
| O7  | Compliance certifications             | SOC 2 Type II attestation                | Report issued by auditor                                                 | 30 September 2028           |
| O8  | GoldBod Act 1140 integration          | Phase 3 GoldBod API integration live     | ≥ 95% of small-scale-channel trades auto-routed through GoldBod workflow | 30 June 2028                |
| O9  | Customer satisfaction                 | NPS (Net Promoter Score)                 | ≥ 40                                                                     | Quarterly from launch       |
| O10 | Unit economics                        | LTV:CAC ratio                            | ≥ 3:1 (target 5:1)                                                       | Measured at month 12        |
| O11 | Financial break-even                  | Cumulative cash position = 0             | Month 22 (Q3 2029)                                                       | 31 July 2029                |

**Table 4.1.** Eleven SMART objectives. O8 (GoldBod integration) is new to this revision, reflecting the April 2025 regulatory transition. O11 break-even date moved from Q4 2028 to Q3 2029 reflecting the revised Q3 2027 launch.

## 4. Scope

Scope is the most important section of any Charter because it is the section that prevents scope creep. The In-Scope list below is the authoritative MVP feature inventory; any feature not listed here is Out-of-Scope and must not be built without a change-control request. The Out-of-Scope list is not exhaustive but enumerates the most likely sources of creep and explicitly defers them.

### 4.1 In-Scope (MVP

#### — Phase 1 + Phase 2)

- **Digital Member Portal (Phase 1):** Member registration; KYC/AML verification via Smile Identity; Minerals Commission licence upload and verification; GoldBod registration verification for small-scale gold traders (per Act 1140, post April 2025); PMMC LGE permit upload; digital member profile; document repository; in-app messaging; notification centre; role-based access (buyer, seller, Chamber approver, administrator).
- **Trading Platform (Phase 2):** Gold lot creation (weight, purity, serial numbers, photos, assay certificate upload); RFQ submission and response; bid comparison dashboard with live LBMA PM Fix overlay; negotiation and counter-offer thread; Chamber trade-approval workflow; digital contract generation and e-signature; escrow integration with partner bank (Stanbic Bank Ghana or Ecobank Ghana); delivery tracking against PMMC release and, where applicable, GoldBod release.
- **Compliance Core (cross-cutting Phase 1+2):** Immutable append-only audit log for every trade state change, pre-validated against the LBMA Responsible Gold Guidance five-step due diligence [lbma.org.uk, 2024]; PII handling per Ghana Data Protection Act 2012 (Act 843) [dataprotection.org.gh]; SOC 2-aligned access controls; audit-package export for GoldBod, PMMC, Minerals Commission, GRA, and LBMA-aligned refiner review.
- **Infrastructure & DevOps:** AWS Ghana region (af-west-1) primary deployment with data residency per DPA 2012 (Act 843); disaster recovery to AWS South Africa (af-south-1); CI/CD pipeline; observability (Datadog, Sentry); on-call rotation with 30-minute P1 acknowledgement SLA.

### 4.2 Out-of-Scope

#### (Deferred to Future Phases)

- **GoldBod API deep integration (small-scale channel automation):** Deferred to Phase 3. MVP supports GoldBod via audit-package export only — the Compliance Core produces the

documentation GoldBod reviewers require, but the API automation is built in Phase 3 (target go-live 30 June 2028, objective O8). This is the largest single Phase 3 work item.

- **PMMC API deep integration (export release automation):** Deferred to Phase 3. MVP supports PMMC LGE workflow via audit-package export; full PMMC API automation in Phase 3 alongside GoldBod.
- **Auction module (timed, reverse, sealed-bid):** Deferred to Phase 3. Corresponds to VPC Gain G5 (sale-timing optimization for high-demand lots) and is explicitly noted in the Golden Thread as a deferred gain.
- **Native mobile apps (iOS / Android):** Deferred to a post-launch Phase 5. The MVP is web-only (responsive, mobile-friendly).
- **Assay service coordination, armored transport booking, vault management, insurance referral:** Deferred to Phase 3 Value-Added Services. The MVP supports upload of externally-produced assay certificates but does not itself request or coordinate assays.
- **Market intelligence dashboard, AI price prediction, fraud/collusion detection, regional demand forecasting, sale-timing recommendations:** Deferred to Phase 4 Analytics. The MVP data model is designed to support these, but the features themselves are not built.
- **White-label versions for other commodity associations (cocoa, timber, diamonds):** Deferred to a future strategic expansion; not budgeted in this Charter.
- **ERP integrations (SAP, Oracle, NetSuite) and bank API marketplaces:** Out-of-Scope. The MVP exposes no public API; Tier 1 Enterprise buyers receive API access only at the Enterprise subscription tier (post-launch).
- **Offline desktop support and offline-first web:** Out-of-Scope. The MVP requires internet connectivity; the regulatory workflow assumes real-time Chamber approval.

## 4.3 Key Deliverables

Table 4.2. Key Deliverables

| Deliverable                     | Format                                            | Owner           | Acceptance Criterion                     |
|---------------------------------|---------------------------------------------------|-----------------|------------------------------------------|
| Working MVP software            | Production deployment on AWS Ghana (af-west-1)    | CTO             | O1–O6 objectives met                     |
| Technical documentation         | README + architecture decision records + runbooks | CTO             | Reviewed and signed off by Lead Engineer |
| Member onboarding documentation | PDF + in-app help centre (English + Twi)          | Head of Product | Reviewed by Chamber + 3 pilot members    |

|                                    |                                                                        |                                           |                                                              |
|------------------------------------|------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------|
| Compliance certification package   | SOC 2 Type II report (year 1)                                          | External auditor                          | Report issued by 30 Sep 2028 (O7)                            |
| GoldBod Act 1140 compliance review | External counsel memo + GoldBod liaison sign-off                       | External counsel + PM                     | Signed off before M3 Development begins                      |
| Training materials                 | Video walkthroughs + live training sessions ×3 (Accra, Kumasi, Tarkwa) | Head of Product                           | ≥ 80% pilot members complete onboarding within 14 days       |
| Source code repository             | GitHub Enterprise, with CI/CD                                          | CTO                                       | All code reviewed; security scan clean                       |
| Audit-log design document          | Architecture decision record pre-validated against LBMA RGG            | Lead Backend Engineer                     | Reviewed by external security auditor + LBMA-aligned auditor |
| Chamber partnership contract       | Signed 5-year revenue-share agreement                                  | Smart Innovations CEO + Chamber President | Signed before MVP launch                                     |

**Table 4.2.** *Nine deliverables (one added: GoldBod Act 1140 compliance review). The Charter is not complete until each deliverable has met its acceptance criterion.*

## 5. Stakeholders & Governance

Governance is the structure that converts a Charter from a planning document into a decision-making authority. The Sponsor is the single point of accountability for budget and scope; the PM is the single point of accountability for delivery. The Steering Committee meets monthly and is the escalation path for any decision above the PM's authority level.

### 5.1

#### Sponsor

- **Name:** 【Sponsor name — to be filled in before sign-off】
- **Title:** Chief Executive Officer, Smart Innovations Ghana Limited
- **Budget authority:** Full authority over the US\$450,000 CapEx ceiling (with the US\$13,450 incremental for GoldBod integration pending Sponsor approval per Section 7.2) and the 18-month OpEx runway; authority to approve changes up to US\$50K cumulative variance without Steering Committee escalation.

## 5.2 Project

### Manager / Lead

- **Name:** 【PM name — to be filled in before sign-off】
- **Title:** Project Manager, AurumXchange Initiative (Smart Innovations Ghana Limited)
- **Authority level:** Authority to expend resources within the US\$463,450 CapEx envelope (assuming Sponsor approves the GoldBod incremental); authority to draw down contingency up to US\$20K without Sponsor approval; authority to make day-to-day scope and prioritization decisions; no authority to approve changes > US\$10K CapEx or > 2 weeks schedule slip.

## 5.3 Key

### Stakeholders & Roles

Table 4.3. Key Stakeholders — RACI Summary

| Role                                           | Name           | Responsibility                                                        | RACI on Major Decisions                               |
|------------------------------------------------|----------------|-----------------------------------------------------------------------|-------------------------------------------------------|
| Sponsor (CEO, Smart Innovations Ghana Limited) | 【Sponsor name】 | Budget authority; ultimate accountability                             | Accountable for budget, scope, Charter sign-off       |
| Project Manager                                | 【PM name】      | Delivery; day-to-day scope & prioritization                           | Responsible for execution; Consulted on scope changes |
| Product Owner / Head of Product                | 【PO name】      | Roadmap; requirements; Chamber liaison                                | Responsible for backlog; Accountable for acceptance   |
| Scrum Master (0.5 FTE)                         | 【SM name】      | Sprint process; impediment removal                                    | Responsible for sprint cadence                        |
| Lead Engineer / Tech Lead                      | 【LE name】      | Architecture; code quality; technical risk                            | Consulted on all technical decisions                  |
| QA Lead                                        | 【QA name】      | Test strategy; quality gate enforcement                               | Responsible for go/no-go on each release              |
| CTO                                            | 【CTO name】     | Technical authority; vendor selection                                 | Accountable for architecture & security               |
| External Counsel (Ghana-barred)                | 【Counsel firm】 | GoldBod Act 1140 compliance review; DPA 2012 compliance; member terms | Consulted on regulatory decisions                     |

|                                             |                                                                                               |                                                         |                                          |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------|
| Chamber Representative (Steering Committee) | Nana Chairperson or delegate                                                                  | Trade rules; member verification; trade approval policy | Consulted on Chamber-affecting decisions |
| End-User Representatives (3 pilot members)  | Abena Owusu (Tarkwa, small-scale), Kofi Mensah (Kumasi, exporter), + 1 Tier 1 (Swiss refiner) | Pilot feedback; UAT                                     | Informed + Consulted on UX decisions     |
| Finance Controller                          | 【FC name】                                                                                     | Variance reporting; accounting treatment                | Responsible for monthly variance report  |

**Table 4.3.** Eleven stakeholders (one added: External Counsel for GoldBod Act 1140 compliance). End-user representatives are now named against real Ghanaian gold-trading locations (Tarkwa, Kumasi) plus a Swiss refiner persona.

## 5.4

### Decision-Making Structure

- **Steering Committee cadence:** Monthly, 90 minutes, first Tuesday of each month. Attendees: Sponsor, PM, CTO, Head of Product, Chamber Representative, Finance Controller, External Counsel (when regulatory items on agenda).
- **Sprint cadence:** Two-week sprints; sprint review on the second Friday; sprint retrospective on the same Friday; backlog grooming on the Wednesday before sprint planning.
- **Escalation path:** Team member → PM (same day) → Sponsor (within 24 hours if budget/scope) → Steering Committee (within 5 business days if > US\$50K or > 2 weeks slip).
- **Chamber trade-approval policy:** Chamber has authority to approve or reject each trade above the platform's auto-approval threshold (initially US\$50,000); the Chamber's trade-approval workflow is integrated into the platform but the approval decision itself is the Chamber's, not Smart Innovations'.
- **GoldBod regulatory liaison:** External Counsel maintains the GoldBod relationship; any change to Act 1140 implementing regulations or the GoldBod API specification triggers immediate Steering Committee notification per Table 3.8.

## 6. High-Level

### Timeline & Milestones

The 13-month build is structured in six phases: Discovery, Design, Development, QA, UAT, and Launch. The sprint cycle is two weeks. Each phase has a single accountable owner and an exit criterion that must be met before the next phase begins; missing a phase exit criterion is a Steering Committee escalation, not a re-planning exercise. The 13-month build (vs the originally assumed 12)



accommodates the GoldBod Act 1140 integration requirement and the SOC 2 readiness engagement, both of which emerged as hard constraints after April 2025.

**Table 4.4. Phase Milestones**

| Phase                 | Window                   | Exit Criterion                                                                                                                                                                           | Owner                              |
|-----------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| M0 — Charter sign-off | By 15 Aug 2026           | Signed Charter + CapEx released (US\$463,450 pending GoldBod incremental approval)                                                                                                       | Sponsor                            |
| M1 — Discovery        | 15 Aug – 30 Sep 2026     | Requirements backlog + technical spike results + Smile Identity KYC integration proof-of-concept + GoldBod Act 1140 compliance review complete                                           | Head of Product + External Counsel |
| M2 — Design           | 1 Oct – 30 Nov 2026      | UX wireframes + audit-log architecture decision record (pre-validated against LBMA RGG) + Chamber sign-off on trade-approval workflow + GoldBod liaison sign-off on integration approach | Head of Product + CTO              |
| M3 — Development      | 1 Dec 2026 – 31 Jul 2027 | Phase 1 + Phase 2 features complete; CI/CD green; sandbox bank escrow integration verified (Stanbic Bank Ghana or Ecobank Ghana)                                                         | CTO                                |
| M4 — QA               | 1 Aug – 31 Aug 2027      | Regression suite passing; pre-launch penetration test complete; 0 critical findings                                                                                                      | QA Lead                            |

|                                                   |                             |                                                                                                                      |                        |
|---------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------|
| M5 — UAT                                          | 1 Sep – 20 Sep 2027         | 3 pilot members (Abena, Kofi, + 1 Tier 1) complete ≥ 1 trade end-to-end; NPS from pilot ≥ 30                         | Head of Product        |
| <b>M6 — Launch</b>                                | 30 September 2027 (Q3 2027) | MVP live in production on AWS Ghana; first real gold trade settled via escrow                                        | PM + CTO               |
| M7 — SOC 2 Type II issued                         | By 30 September 2028        | Auditor report issued                                                                                                | External auditor + CTO |
| M8 — Phase 3 GoldBod + PMMC deep integration live | By 30 June 2028             | ≥ 95% of small-scale-channel trades auto-routed through GoldBod workflow; PMMC release workflow automated end-to-end | CTO + Head of Product  |

**Table 4.4.** *Nine milestones. M6 Launch (30 Sep 2027) is the focal date. M8 (GoldBod + PMMC Phase 3 integration) is the new milestone added in this revision.*

Date reconciliation note: the locked assumption was 'MVP launch Q1 2027.' On a 13-month build from an August 2026 Charter sign-off, the realistic launch window is Q3 2027 (30 September 2027). The original Q1 2027 target would have required an aggressive 6-month build, which is incompatible with the regulated nature of the platform — extended KYC integration, escrow bank onboarding, SOC 2 readiness, and (since April 2025) the GoldBod Act 1140 compliance review all require calendar time, not just engineering effort. The Charter therefore adopts Q3 2027 (30 September 2027) as the contractual launch date; the Financial Documentation's break-even at month 22 shifts accordingly to Q3 2029 (31 July 2029), not Q4 2028 as stated in the prior version of Table 3.7. This is the single reconciliation the Golden Thread must absorb — see the Golden Thread Validation section for the explicit reconciliation narrative.

## 7.

## Resources

### 7.1 Team

#### Composition

- **12 FTE at peak build (December 2026 – July 2027):** 1 CTO (0.5 FTE), 1 Head of Product (1.0 FTE), 4 senior backend engineers (2.0 FTE during peak, 1.0 FTE earlier), 2 senior frontend engineers, 1

DevOps/SRE, 1 QA Lead (joined month 8), 1 UI/UX Designer (0.5 FTE × 6 months), 1 PM (0.5 FTE). Team based in Accra with a satellite presence in Kumasi for member proximity.

- **Post-launch (from October 2027):** Retain 3 FTE for enhancements and incident response (1 backend, 1 frontend, 1 DevOps); add 2 Tier 1 support + 1 Customer Success Manager for a total of 6 FTE in steady-state operations. Engineering team scales back up to 8 FTE for Phase 3 build (Q1 2028), including GoldBod and PMMC API integration specialists.

## 7.2 Estimated

### Budget Summary

- **CapEx (build, 13 months):** US\$463,450 (per Table 3.1) — US\$13,450 (2.9%) above the US\$450,000 ceiling. The overrun is the GoldBod Act 1140 incremental (US\$8,500 legal compliance review + US\$4,000 extra integration prep). Sponsor may either (a) approve a US\$13,450 ceiling increase to US\$463,450, or (b) absorb the overrun from contingency (reducing contingency from US\$60,450 to US\$47,000). Recommended path: (a) ceiling increase, preserving full contingency for unknowns.
- **OpEx runway (18 months post-launch):** Approximately US\$715,000 (per Financial Doc Section B).
- **Total capital required to break-even:** Approximately US\$1,178,450 (US\$463,450 CapEx + US\$715,000 OpEx), plus a US\$170,000 bridge between month 18 (runway end, Q1 2029) and month 22 (break-even, Q3 2029) — to be covered by a Series A round in month 18 (Q1 2029).

## 7.3 Tools,

### Infrastructure & Third-Party Dependencies

- Source control & CI/CD: GitHub Enterprise, GitHub Actions; deployment via ArgoCD or AWS CodePipeline.
- Project management: Jira + Confluence; sprint board, retros, and decision records all in Confluence.
- Observability: Datadog (APM, logs, dashboards), Sentry (error tracking), Snyk (dependency security), GitHub Dependabot.
- Cloud: AWS Ghana region (af-west-1, GA since 2021) for data residency per Ghana DPA 2012 (Act 843); DR to AWS af-south-1 (Cape Town).
- Identity & KYC: Smile Identity (preferred; operates in 50+ African countries per their public positioning [Smile Identity, 2024]) or Youverify (fallback) for KYC/AML at onboarding and ongoing.
- Escrow banking: Stanbic Bank Ghana or Ecobank Ghana — both confirmed operating corporate banking in Ghana [stanbicbank.com.gh; ecobank.com]; selection completed during M1 Discovery; partner must be LBMA-aligned.
- LBMA price feed: LBMA PM Fix via authorized data vendor (FastMarkets or Refinitiv).

- GoldBod (Act 1140): integration with the Ghana Gold Board's small-scale-mined gold channel; regulatory liaison maintained by External Counsel; API integration deep-build deferred to Phase 3 (M8).
- PMMC and Minerals Commission: integration via API where available, manual workflow where not (Phase 3 closes the loop).
- Bank of Ghana: DGPP reference prices (the BoG purchased 358,218 oz from Chamber members in 2024 [Ghana Chamber of Mines PWYP report, 2024]) and forex verification for export proceeds.
- External auditor for SOC 2 Type II: Selection completed during M1 Discovery; contract signed before M3 begins.

## 8. Risks, Assumptions, Constraints & Mitigations

### 8.1 Known Risks & Mitigations

Table 4.5. Risk Register (Top 11)

| #  | Risk                                                                                       | Category   | Likelihood | Impact | Mitigation                                                                                                                                                                                        |
|----|--------------------------------------------------------------------------------------------|------------|------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | GoldBod Act 1140 implementing regulations or API specification changes during build window | Regulatory | High       | High   | External Counsel maintains GoldBod relationship; quarterly regulatory review; sandbox integration proof-of-concept in M1 Discovery to surface spec changes early; 15% contingency absorbs rework. |

|    |                                                                    |             |        |           |                                                                                                                                             |
|----|--------------------------------------------------------------------|-------------|--------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| R2 | KYC/AML API latency or downtime during onboarding (Smile Identity) | Technical   | Medium | High      | Fallback to Youverify; queue onboarding if both fail; SLA of 30s p95 with Smile Identity in contract.                                       |
| R3 | Escrow bank onboarding takes > 90 days (Stanbic or Ecobank)        | Operational | Medium | High      | Begin bank selection during M1 Discovery (Aug-Sep 2026); pre-negotiate with 2 candidate banks; fallback to a manual escrow workflow in MVP. |
| R4 | PMMC API not available for export workflow integration             | Regulatory  | High   | Medium    | MVP delivers audit-package export only; full PMMC API integration deferred to Phase 3 (M8) with manual workaround in interim.               |
| R5 | LBMA Responsible Gold Guidance audit fails on platform design      | Compliance  | Low    | Very High | Engage LBMA-aligned auditor in M2 Design; pre-validate audit-log architecture against LBMA RGG five-step due diligence [lbma.org.uk, 2024]. |

|    |                                                                                                 |        |        |      |                                                                                                                                                                                                                            |
|----|-------------------------------------------------------------------------------------------------|--------|--------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R6 | Chamber member adoption below projection (< 250 verified by Y1 end)                             | Market | Medium | High | Co-funded marketing with Chamber; 14-day free Professional trial; in-person demo events in Accra, Kumasi, Tarkwa, Obuasi; PM triggers revenue-model refresh if Q2 actuals > 20% below plan.                                |
| R7 | Competitor launches a Ghana-region gold trading platform (incl. GoldBod's own digital platform) | Market | Medium | High | 5-year exclusive revenue-share contract with Chamber is the moat. Maintain feature velocity via Phase 3–4 roadmap; if GoldBod launches its own platform, position AurumXchange as the Chamber-member-governed alternative. |

|     |                                                                        |             |        |        |                                                                                                                                                               |
|-----|------------------------------------------------------------------------|-------------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R8  | Key person dependency (CTO or Lead Backend Engineer leaves)            | Operational | Medium | High   | Document architecture decision records; pair-programming on audit-log module; key-person insurance for CTO; competitive retention package.                    |
| R9  | AWS Ghana region (af-west-1) availability degraded during build window | Technical   | Low    | Medium | Fallback to AWS af-south-1 (Cape Town) with contractual data-residency terms; revisit af-west-1 when service stabilizes.                                      |
| R10 | 0.10% transaction fee rejected by Tier 1 buyers as too high            | Market      | Low    | Medium | Hybrid fee structure already includes cap-at-maximum option; Enterprise tier negotiates volume-based discounts; PM reviews fee sensitivity in Q1 post-launch. |

|     |                                                                              |            |     |        |                                                                                                                                                                  |
|-----|------------------------------------------------------------------------------|------------|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R11 | Ghana AML Act 2020 (Act 1044) amendment increases KYC requirements mid-build | Regulatory | Low | Medium | Maintain quarterly legal review with external counsel; design KYC module to be configurable rather than hard-coded; FIC liaison maintained via External Counsel. |
|-----|------------------------------------------------------------------------------|------------|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Table 4.5.** Top 11 risks (one added: R1 on GoldBod Act 1140 regulatory change — now the highest-impact regulatory risk). Risk register is maintained live in Confluence; new risks are added as discovered.

## 8.2

### Assumptions

The following assumptions must hold for this Charter and the supporting Financial Documentation to be valid. If any assumption is invalidated, the Charter must be re-baselined. Each assumption is grounded in a publicly verifiable source where possible.

- The Ghana Chamber of Gold Buyers signs the 5-year exclusive revenue-sharing agreement before MVP launch.
- At least one LBMA-aligned Ghana-licensed bank (Stanbic Bank Ghana or Ecobank Ghana) agrees to provide escrow services within 90 days of Charter sign-off.
- Smile Identity confirms Ghana KYC API availability with 30-second p95 latency [Smile Identity, 2024 — operates in 50+ African countries].
- AWS Ghana region (af-west-1, GA since 2021) remains generally available throughout the build window.
- Ghana's 2024 gold export revenue of US\$11.64B and 2024 production of 4.9M ounces are accurate and representative of the addressable market [CGTN Africa, 2025; trade.gov, June 2025; Reuters, May 2025].
- The Chamber's 1,500 licensed members is an accurate figure (verifiable via Chamber registry).
- 17% of Chamber members adopt the platform within 12 months of launch (250 verified members by Y1 end).
- Average trade value of US\$120,000 ( $\approx$  1.8 kg at the late-2024 LBMA PM Fix of  $\sim$ US\$2,600/oz [LBMA, 2024]) is representative of Chamber-member trades.
- Annual churn of 12% blended (15% buyer / 8% seller) is achievable.



- The 18-month OpEx runway of approximately US\$715,000 is funded by Smart Innovations Ghana Limited's balance sheet.
- A Series A bridge round of US\$500K closes by month 18 (Q1 2029) to cover months 18–22 (the gap between runway end and break-even).
- The GoldBod Act 1140 implementing regulations and API specification remain stable through M3 Development; if they change, contingency and the GoldBod-specific change-control mechanism (Table 3.8) absorb the impact.

### 8.3 Regulatory & Compliance Constraints

- **Ghana Gold Board Act, 2025 (Act 1140):** GoldBod is the sole buyer, assayer, and exporter of small-scale-mined gold since 2 April 2025 [UNCTAD Investment Policy Monitor, 2025; goldbod.gov.gh]. AurumXchange must support GoldBod verification for any trade touching small-scale-channel gold; deep API integration in Phase 3 (M8).
- **Ghana AML Act, 2020 (Act 1044):** Establishes the Financial Intelligence Centre (FIC) and places AurumXchange within DNFBPs [OGP commitment GH0029; namescan.io AML Regulations Ghana, 2025]. Platform must support ongoing monitoring and suspicious-activity reporting to the FIC.
- **Ghana Data Protection Act, 2012 (Act 843):** PII handling, data subject rights, breach notification within 72 hours; data residency in Ghana (AWS af-west-1). Enforced by the Data Protection Commission [dataprotection.org.gh].
- **Minerals Commission of Ghana licensing:** Every seller must hold a valid licence [mincom.gov.gh]; platform must verify and re-verify monthly.
- **PMMC export certification:** Every trade intended for export must be PMMC-released by a Licensed Gold Exporter (LGE); MVP supports audit-package export only, full PMMC API integration in Phase 3 (M8) [pmmc.gov.gh].
- **LBMA Responsible Gold Guidance (RGG):** Five-step due diligence framework on OECD DDG for Conflict-Affected and High-Risk Areas [lbma.org.uk, 2024]. Audit trail must support responsible-sourcing attestation; LBMA-aligned auditor engaged in M2 Design.
- **OECD Due Diligence Guidance:** Chain-of-custody documentation for international buyers; audit-log architecture designed to satisfy.
- **SOC 2 Type II:** Annual attestation by external auditor; report issued by 30 September 2028 (O7).
- **ISO 27001 ISMS:** Certification pursued in Year 2 post-launch.
- **PCI DSS Level 1:** Required for escrow and payment integration; achieved via partner bank's PCI scope (Smart Innovations does not directly handle PANs).
- **GDPR equivalence:** For international Tier 1 buyers' data; treated under Ghana DPA 2012 with contractual GDPR-equivalence provisions in member terms.

## 9. Success

### Metrics (KPIs)

Success is defined as the achievement of the SMART objectives in Section 3, measured against the KPIs below. A green status requires all Critical KPIs to be on target; amber allows one Critical KPI to be off-target with a remediation plan; red triggers Steering Committee intervention. The KPI dashboard is reviewed monthly at the Steering Committee meeting.

**Table 4.6. Success Metrics (KPIs)**

| KPI                                       | Category       | Target                                                                     | Measurement Cadence         |
|-------------------------------------------|----------------|----------------------------------------------------------------------------|-----------------------------|
| Verified member count                     | Adoption       | 250 by Y1 end (Sep 2028); 1,050 by Y5 end (Sep 2032)                       | Monthly                     |
| Trade volume (US\$M)                      | Adoption       | ≥ US\$350M Y1; ≥ US\$1,822M Y5                                             | Monthly                     |
| System uptime                             | Reliability    | ≥ 99.9% monthly                                                            | Continuous                  |
| P95 API response time                     | Performance    | < 500ms for all critical endpoints                                         | Continuous                  |
| Critical-severity security findings       | Security       | 0 at launch; ≤ 2 per quarter (with remediation)                            | Per release + quarterly     |
| SOC 2 Type II attestation                 | Compliance     | Report issued by 30 Sep 2028                                               | Annual                      |
| GoldBod Act 1140 Phase 3 integration live | Compliance     | ≥ 95% of small-scale-channel trades auto-routed via GoldBod by 30 Jun 2028 | Monthly from Phase 3 launch |
| NPS (Net Promoter Score)                  | Satisfaction   | ≥ 40 (Year 1); ≥ 50 (Year 3)                                               | Quarterly                   |
| LTV:CAC ratio                             | Unit economics | ≥ 3:1 (Year 1); ≥ 5:1 (Year 3)                                             | Quarterly                   |
| Annual churn (blended)                    | Retention      | ≤ 15%                                                                      | Quarterly                   |
| Net Revenue Retention                     | Retention      | ≥ 100% (Year 1); ≥ 110% (Year 3)                                           | Quarterly                   |
| CAC payback period                        | Unit economics | ≤ 12 months                                                                | Quarterly                   |

|                                                                                  |           |                                                                     |            |
|----------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------|------------|
| Break-even date                                                                  | Financial | Month 22 (Q3 2029, given Q3 2027 launch)                            | Continuous |
| Compliance incidents (any of: SOC 2, ISO 27001, DPA 2012, GoldBod, AML Act 1044) | Risk      | 0 per quarter (target); any incident → 24-hour sponsor notification | Continuous |

**Table 4.6.** Fourteen KPIs (two added: GoldBod Phase 3 integration; compliance incident scope now explicitly includes GoldBod and AML Act 1044).

## 9.1 Definition of

### Done

The MVP is 'Done' when: (i) all Phase 1 + Phase 2 features are deployed to production on AWS Ghana (af-west-1); (ii) the first real gold trade has been settled end-to-end through the platform's escrow workflow with Stanbic Bank Ghana or Ecobank Ghana; (iii) the pre-launch penetration test reports 0 critical-severity findings; (iv) at least 3 pilot members (Abena Owusu, Kofi Mensah, + 1 Tier 1) have completed UAT with NPS  $\geq 30$ ; (v) the SOC 2 Type II readiness engagement is complete and the year-1 attestation window is open; (vi) the Chamber partnership contract is signed; (vii) the audit-log design document has been reviewed by the external security auditor and pre-validated against the LBMA Responsible Gold Guidance five-step due diligence; and (viii) the GoldBod Act 1140 compliance review (External Counsel memo) is signed off. Items (i)–(viii) must all be complete before the Sponsor signs the Launch Authorization.

## 10.

### Authorization

By signing below, the Sponsor authorizes the Project Manager to expend resources within the scope and budget defined in this Charter, including the US\$463,450 CapEx budget (Table 3.1, pending Sponsor approval of the US\$13,450 GoldBod-related incremental above the US\$450,000 ceiling) and the 18-month OpEx runway of approximately US\$715,000 (Financial Doc Section B). The Sponsor grants the PM authority to make day-to-day scope and prioritization decisions, to draw down contingency up to US\$20K without further approval, and to escalate to the Sponsor for any change exceeding the thresholds defined in Table 3.8. This Charter is binding upon signature and remains in force until the project is formally closed by the Sponsor or until a successor Charter is signed.

### Sponsor Authorization:

| Field | Value          |
|-------|----------------|
| Name  | 【Sponsor name】 |

|                             |                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                       | Chief Executive Officer, Smart Innovations Ghana Limited                                                                                                                                                                                                                                                  |
| Signature                   | _____                                                                                                                                                                                                                                                                                                     |
| Date                        | ____ / ____ / 2026                                                                                                                                                                                                                                                                                        |
| Explicit grant of authority | The PM is hereby authorized to expend resources within the US\$463,450 CapEx envelope (pending Sponsor approval of the US\$13,450 GoldBod-related incremental above the US\$450,000 ceiling) and the 18-month OpEx runway defined in this Charter, subject to the change-control thresholds in Table 3.8. |

**Project Manager Acceptance:**

| Field                | Value                                                                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                 | 【PM name】                                                                                                                                                                       |
| Title                | Project Manager, AurumXchange Initiative (Smart Innovations Ghana Limited)                                                                                                      |
| Signature            | _____                                                                                                                                                                           |
| Date                 | ____ / ____ / 2026                                                                                                                                                              |
| Acceptance statement | I accept the scope, authority, and accountability defined in this Charter and commit to delivering the SMART objectives in Section 3 within the budget envelope in Section 7.2. |

**Chamber Acknowledgement (counterparty):**

| Field     | Value                                   |
|-----------|-----------------------------------------|
| Name      | 【Chamber President name】                |
| Title     | President, Ghana Chamber of Gold Buyers |
| Signature | _____                                   |

|                 |                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date            | ____ / ____ / 2026                                                                                                                                                                                                                        |
| Acknowledgement | The Chamber acknowledges the scope and commercial model defined in this Charter and commits to the responsibilities allocated to the Chamber in Table 2.4 (BMC Key Partnerships) and the partnership agreement referenced in Section 7.3. |